



Generative AI Applications in SE Lab 04

Submitted to

Dr. Humaira Waqas

Submitted by

M Ehtisham Amjad (231996)

BSSE-VI-B

Graded Task 1: Relevance Score Threshold Filter

Objective

Implement a minimum relevance score threshold (≥ 2) in the Retrieval Code node to filter low-quality matches and distinguish between high-confidence, low-confidence, and no-context scenarios.

Implementation

The Retrieval Code node was updated with:

- Score threshold logic: only entries with score ≥ 2 included in context
- Three-way status classification: 'success' (score ≥ 2), 'low_confidence' (score < 2), 'no_context' (no matches)
- Added 'topScore' field to response JSON for client-side confidence display

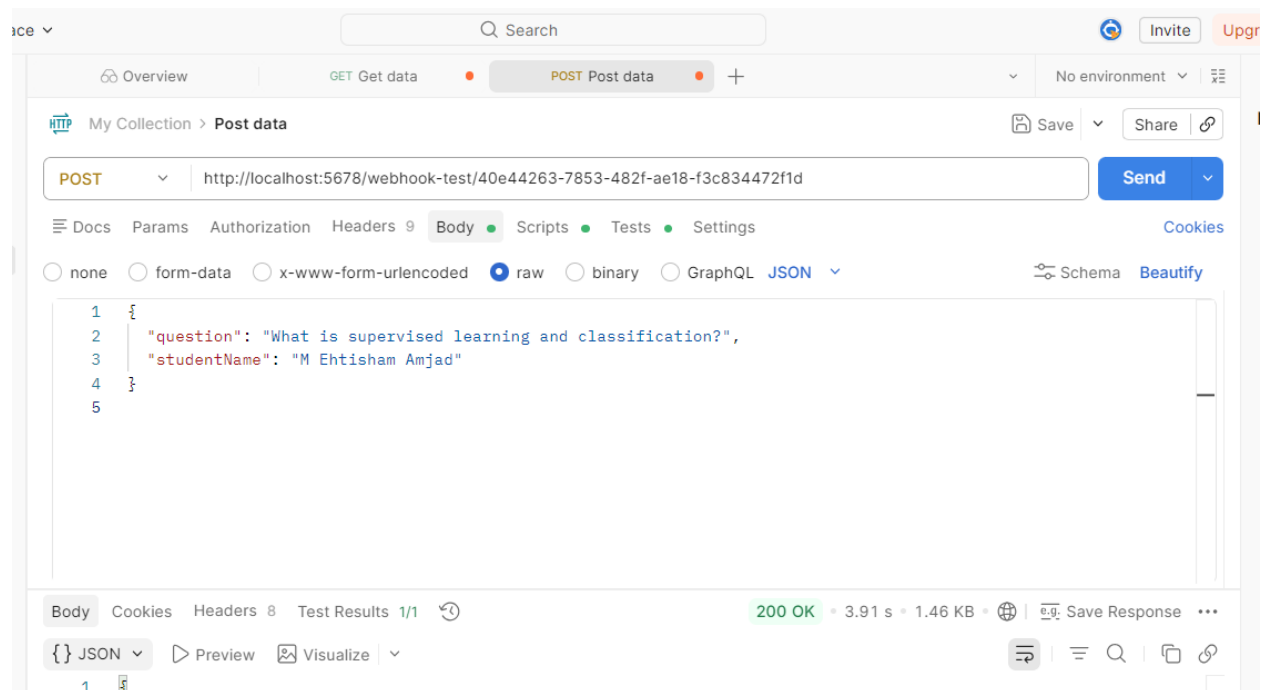
Test Case 1 — High-Confidence Match

Request sent:

```
{
  "question": "What is supervised learning and classification?",
  "studentName": "M Ehtisham Amjad"
}
```

Expected Result:

200 Success with topScore ≥ 2 , context block populated, AI answer grounded in KB content. The sources field references KB-001. The first IF node routes to TRUE (output 0) triggering the LLM chain.



Body Cookies Headers 8 Test Results 1/1 200 OK • 3.91 s • 1.46 KB • Save Response

```

1 {
2   "status": "success",
3   "studentName": "M Ehtisham Amjad",
4   "question": "What is supervised learning and classification?",
5   "answer": "Supervised learning involves training a model using labelled data to predict specific outcomes.
6             Classification is a type of supervised learning where the model predicts discrete categories. For
7             example, in fraud detection, classification algorithms are trained on historical labelled transaction
8             data to differentiate between fraudulent and legitimate activities.\n\nSources:\n\nMitchell, T. (1997).
9             Machine Learning.\n\nBhattacharyya et al. (2011). Data Mining for Credit Card Fraud. Decision Support
10            Systems.",
11  "sources": "KB-001 – Machine Learning (Mitchell, T. (1997). Machine Learning.); KB-002 – Machine Learning
12            (Goodfellow et al. (2016). Deep Learning.); KB-009 – Financial Fraud Detection (Bhattacharyya et al.
13            (2011). Data Mining for Credit Card Fraud. Decision Support Systems.)",
14  "retrievedDocs": 3,
15  "contextUsed": true,
16  "timestamp": "2026-03-29T13:06:09.921-04:00"
17 }
```

Test Case 2 — Low-Confidence Match

Request sent:

```

{
  "question": "What is AI coding?",
  "studentName": "M Ehtisham Amjad"
}
```

Expected Result:

200 Success with status: 'low_confidence', topScore = 1, fallback message returned. LLM is NOT called. The first IF node routes FALSE, the second IF node routes TRUE (lowConfidence branch).

ice ▾ Search Invite Up

Overview GET Get data POST Post data No environment

HTTP My Collection > Post data Save Share

POST http://localhost:5678/webhook-test/40e44263-7853-482f-ae18-f3c834472f1d Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```

1 {
2   "question": "What is AI coding?",
3   "studentName": "M Ehtisham Amjad"
4 }
5
```

Body Cookies Headers 8 Test Results 1/1 200 OK • 1.75 s • 844 B • Save Response

```

1 {
2   "status": "low_confidence",
3   "studentName": "M Ehtisham Amjad",
4   "question": "What is AI coding?",
5   "answer": "I found some related information but it may not fully answer your question. Please consult your
6     course materials for more details.",
7   "sources": [],
8   "retrievedDocs": 0,
9   "topScore": 1,
10  "contextUsed": false,
11  "timestamp": "2026-03-29T13:07:30.661-04:00"
12 }

```

Test Case 3 — No Context Found

Request sent:

```

{
  "question": "What are the rules of cricket?",
  "studentName": "M Ehtisham Amjad"
}

```

Expected Result:

200 Success with status: 'no_context', topScore: 0, empty sources array. LLM is NOT called. Both IF nodes route FALSE and the fallback Respond to Webhook1 node executes.

```

1 {
2   "question": "What are the rules of cricket?",
3   "studentName": "M Ehtisham Amjad"
4 }
5

```

```

1 {
2   "status": "low_confidence",
3   "studentName": "M Ehtisham Amjad",
4   "question": "What are the rules of cricket?",
5   "answer": "I found some related information but it may not fully answer your question. Please consult your
6     course materials for more details.",
7   "sources": [],
8   "retrievedDocs": 0,
9   "topScore": 1,
10  "contextUsed": false,
11  "timestamp": "2026-03-29T13:08:56.228-04:00"
12 }

```

Graded Task 2: Multi-Topic Knowledge Base Expansion

Objective

Expand the knowledge base with 6+ new entries covering a domain relevant to your research. Each entry must have ≥ 5 keywords and ≥ 3 sentences of factual content.

Knowledge Base Entries

Topic: Financial Systems (Fraud Detection, Credit Scoring, AML, Payment Systems, Algorithmic Trading, Financial Regulation)

Google Sheet Link:

<https://docs.google.com/spreadsheets/d/12iEbE5upONqQ91Cie20ODLnwMhH3ErWY00U3CFVe00E/edit?gid=0#gid=0>

Test Cases

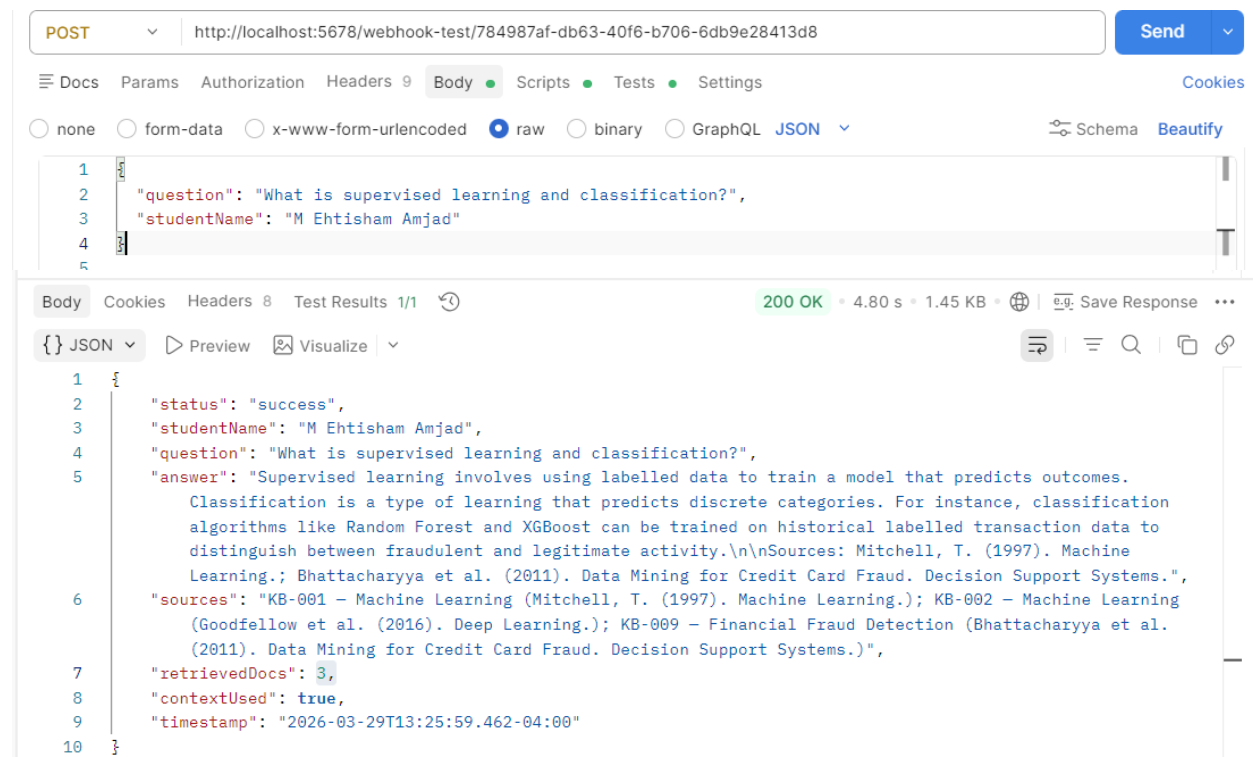
Test 1 — Retrieval Success: Financial Fraud Detection

Request sent:

```
{
  "question": "What is supervised learning and classification?",
  "studentName": "M Ehtisham Amjad"
}
```

Expected Result:

Retrieves KB-001, KB-002, and KB-009. Status: "success", contextUsed: true, retrievedDocs: 3. Sources field references KB-001 (Machine Learning), KB-002 (Machine Learning), and KB-009 (Financial Fraud Detection). AI answer synthesises supervised learning concepts with fraud classification use case.



Test 2 — Retrieval Success: Credit Scoring

Request sent:

```
{
  "question": "How does logistic regression work in credit risk assessment?",
  "studentName": "M Ehtisham Amjad"
}
```

Expected Result:

Retrieves KB-010. Status: "success", contextUsed: true. Sources field references KB-010 (Credit Scoring, Thomas et al. 2002). AI answer explains logistic regression, default prediction, and creditworthiness assessment grounded in KB content.

The screenshot displays a REST client interface with the following details:

- URL:** `http://localhost:5678/webhook-test/784987af-db63-40f6-b706-6db9e28413d8`
- Method:** POST
- Body (raw JSON):**

```
{
  "question": "How does logistic regression work in credit risk assessment?",
  "studentName": "M Ehtisham Amjad"
}
```
- Status:** 200 OK
- Response Body (JSON):**

```
{
  "status": "success",
  "studentName": "M Ehtisham Amjad",
  "question": "How does logistic regression work in credit risk assessment?",
  "answer": "Logistic regression is utilized within credit scoring models to assess the likelihood of a borrower defaulting on a loan. These models heavily weigh features such as payment history, debt-to-income ratio, and credit utilisation. Through this assessment, a low credit score indicates a high default risk, potentially leading to loan rejection or higher interest rates.\n\nSources: Thomas et al. (2002). Credit Scoring and Its Applications. SIAM.",
  "sources": "KB-010 – Credit Scoring (Thomas et al. (2002). Credit Scoring and Its Applications. SIAM.); KB-013 – Algorithmic Trading Risk (Aldridge, I. (2013). High-Frequency Trading. Wiley Finance.); KB-014 – Financial Regulation (Basel Committee on Banking Supervision (2017). Basel III: Finalising Post-Crisis Reforms. BIS.)",
  "retrievedDocs": 3,
  "contextUsed": true,
  "timestamp": "2026-03-29T13:43:18.974-04:00"
}
```

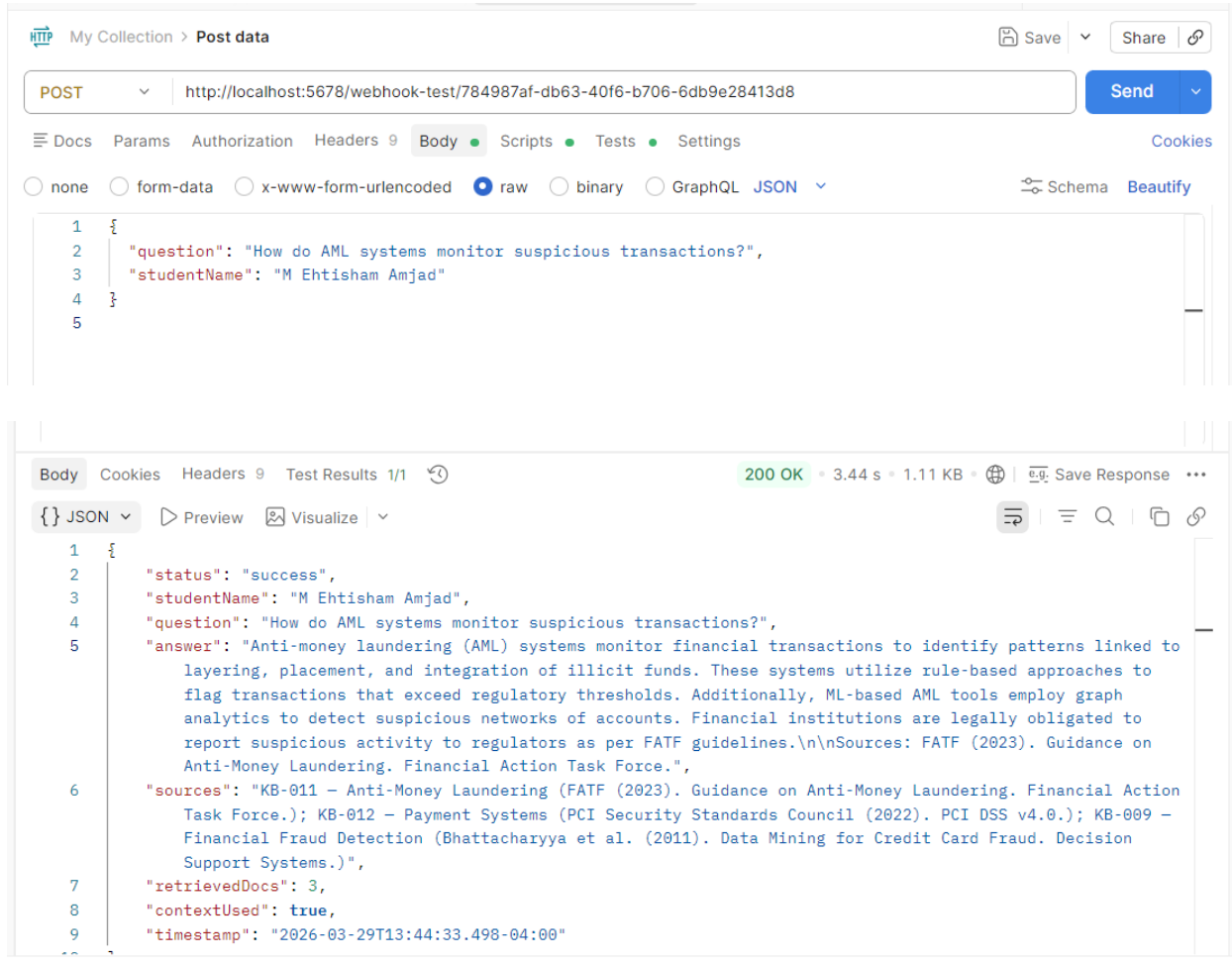
Test 3 — Retrieval Success: Anti-Money Laundering

Request sent:

```
{
  "question": "How do AML systems monitor suspicious transactions?",
  "studentName": "M Ehtisham Amjad"
}
```

Expected Result:

Retrieves KB-011. Status: "success", contextUsed: true. Sources field references KB-011 (Anti-Money Laundering, FATF 2023). AI answer explains transaction monitoring, suspicious activity detection, and compliance grounded in KB content.

**Graded Task 3: RAG Audit Log & Research Reflection****Objective**

Implement a post-response audit logging system that records successful RAG queries to a Google Sheet. Write a 1-page research reflection on RAG applications, limitations, and production requirements.

Implementation

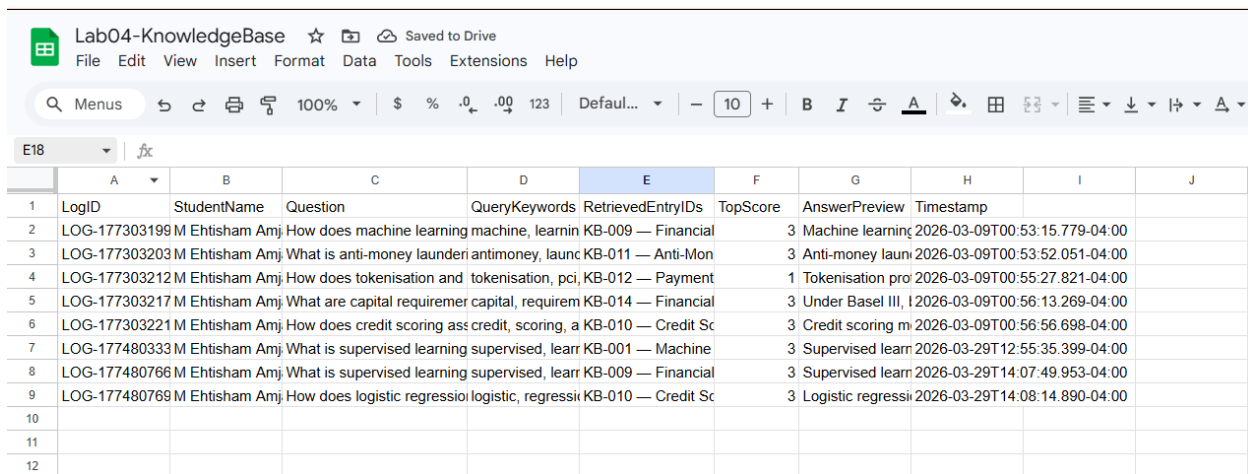
Added a Google Sheets 'Append Row' node after the Respond to Webhook node (execute after response pattern) targeting the RAGAuditLog tab in the same spreadsheet. The node appends one row per successful query with the following columns:

- LogID — unique identifier generated as "LOG-" + current timestamp in milliseconds
- StudentName — M Ehtisham Amjad

- Question — the user's query from the webhook body
- QueryKeywords — extracted search terms output by the Query Prep Code node
- RetrievedEntryIDs — semicolon-separated list of matched KB entry IDs and topics
- TopScore — highest relevance score from the retrieval step
- AnswerPreview — first 100 characters of the AI-generated answer
- Timestamp — ISO 8601 formatted timestamp of the query

RAG Audit Log Sheet Link:

<https://docs.google.com/spreadsheets/d/12iEbE5upONqQ91Cie20ODLnwMhH3ErWY00U3CFVe0OE/edit?gid=736941132#gid=736941132>



LogID	StudentName	Question	QueryKeywords	RetrievedEntryIDs	TopScore	AnswerPreview	Timestamp
LOG-177303199	M Ehtisham Amj	How does machine learning machine, learnin	KB-009 — Financial		3	Machine learning	2026-03-09T00:53:15.779-04:00
LOG-177303203	M Ehtisham Amj	What is anti-money launderi antimoney, launc	KB-011 — Anti-Mon		3	Anti-money laun	2026-03-09T00:53:52.051-04:00
LOG-177303212	M Ehtisham Amj	How does tokenisation and tokenisation, pci, KB-012 — Payment			1	Tokenisation pro	2026-03-09T00:55:27.821-04:00
LOG-177303217	M Ehtisham Amj	What are capital requiremer capital, requirem	KB-014 — Financial		3	Under Basel III, I	2026-03-09T00:56:13.269-04:00
LOG-177303221	M Ehtisham Amj	How does credit scoring ass credit, scoring, a	KB-010 — Credit Sc		3	Credit scoring m	2026-03-09T00:56:56.698-04:00
LOG-177480333	M Ehtisham Amj	What is supervised learning supervised, learr	KB-001 — Machine		3	Supervised learn	2026-03-29T12:55:35.399-04:00
LOG-177480766	M Ehtisham Amj	What is supervised learning supervised, learr	KB-009 — Financial		3	Supervised learn	2026-03-29T14:07:49.953-04:00
LOG-177480769	M Ehtisham Amj	How does logistic regressioni logistic, regressi	KB-010 — Credit Sc		3	Logistic regressi	2026-03-29T14:08:14.890-04:00

Research Reflection

1. RAG Application in My Research Domain

My research interest is in Financial AI and automated compliance systems. RAG would directly enable an intelligent financial advisory assistant that answers analyst queries using verified regulatory documents and internal risk reports. The knowledge base would contain structured content including Basel III guidelines, FATF AML directives, PCI DSS requirements, internal credit risk policies, and historical audit reports chunked by regulation and section. When a compliance officer or risk analyst submits a question, the retrieval component fetches the most relevant regulatory chunks and injects them as context into the LLM prompt, grounding the response in authoritative documents rather than general knowledge. This is critical in financial services where hallucinated regulatory guidance could result in legal liability. The knowledge base can be updated as regulations change without any model retraining, making it practical for compliance teams to maintain. This architecture mirrors systems already deployed by financial institutions for regulatory Q&A and is directly applicable to research in FinTech, RegTech, and AI-driven risk management.

2. Three Key Limitations of Keyword-Based Retrieval vs Vector Search

Limitation 1: Synonym blindness — keyword search fails when the query uses different but equivalent terms. A compliance officer asking 'What are the capital buffer requirements?' will not

match a KB entry that uses 'minimum capital ratios' and 'capital adequacy' because the keywords do not overlap. Vector search maps both query and KB entries to embeddings where semantically equivalent expressions are located near each other in the vector space, retrieving the correct entry regardless of exact word choice.

Limitation 2: No semantic understanding of intent — the scoring algorithm counts raw keyword occurrences so a long KB entry mentioning many common financial terms scores higher even if it is only loosely relevant to the query. Vector search uses cosine similarity in a high-dimensional embedding space which measures conceptual alignment rather than term frequency, producing more precise relevance rankings particularly for technical regulatory queries.

Limitation 3: Poor handling of multi-word regulatory concepts — a query like 'What triggers a suspicious activity report?' produces individual keywords that may partially match unrelated entries. Vector embeddings capture the full semantic intent of the phrase as a unit, allowing retrieval of the correct AML entry even when individual keywords overlap with unrelated financial entries in the knowledge base.

3. Production Requirements Missing from Lab Implementation

The lab implementation demonstrates the core RAG architecture but omits several critical production requirements. Authentication and authorisation are absent — any client with the webhook URL can query the system; a production financial compliance assistant would require OAuth2 or API key validation with role-based access control to restrict which regulatory documents different user roles can query. The Google Sheets backend is unsuitable for production — every request reads the entire sheet with no indexing, causing performance degradation as the KB grows; a production system would use a vector database such as pgvector or Pinecone with approximate nearest neighbour search. There is no rate limiting, audit trail for access control, or retry logic for failed LLM calls. The single-row chunking strategy is insufficient for long regulatory documents which may span dozens of pages; production RAG requires overlapping chunk windows with approximately 512 tokens and 50-token overlap to preserve context across boundaries. Finally there is no model versioning, answer quality monitoring, or human-in-the-loop review mechanism — all of which are mandatory in regulated financial environments where AI-generated guidance must be verifiable and auditable.

Workflow Export & Screenshots



Figure 1: Lab04 Task 1 Keyword-Based RAG Pipeline with Relevance Score Threshold and Low-Confidence Detection in n8n

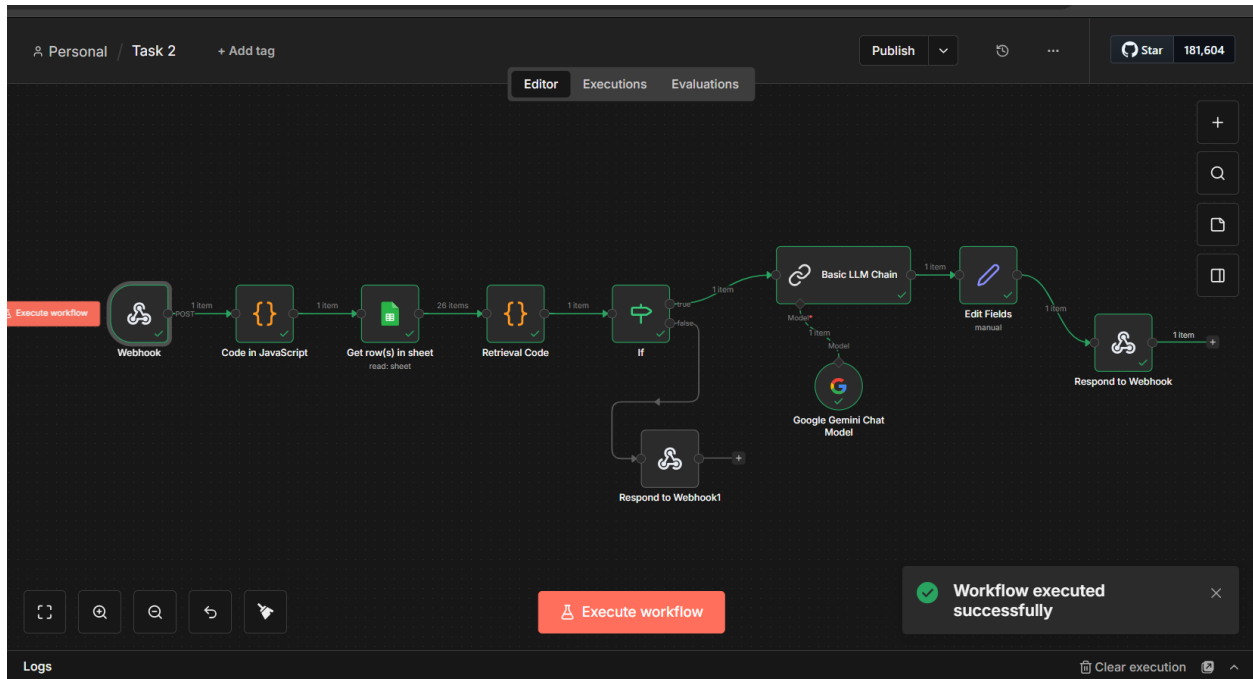


Figure 2: Lab04 Task 2 Multi-Turn RAG Workflow with Conversation History Support in n8n

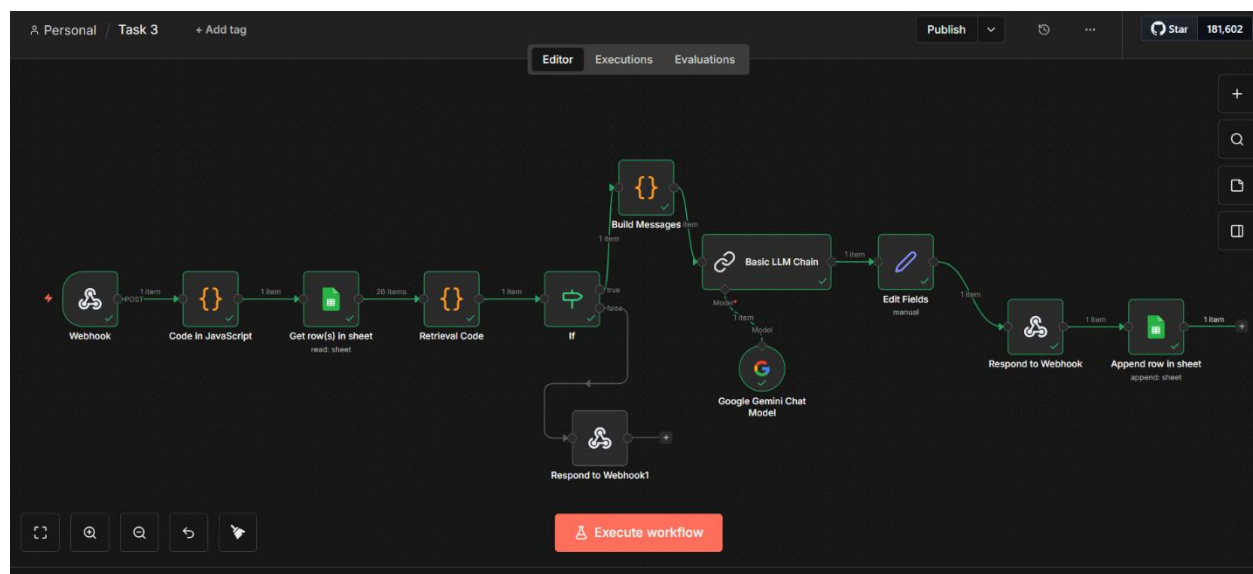


Figure 3: Lab04 Task 3 RAG Pipeline with Audit Logging Workflow in n8n

Known Limitations

- **Keyword-based retrieval:** Cannot understand semantic similarity — query 'ML algorithms' won't match 'machine learning methods' even though they're conceptually identical.
- **No authentication:** The webhook endpoint is public — anyone with the URL can query the knowledge base without authorization.
- **Sheet-based storage:** Google Sheets is not a database — every request reads the entire sheet, causing performance degradation as KB grows. No indexing or query optimization.
- **No caching:** Identical queries trigger full retrieval + LLM generation each time, increasing latency and API costs.
- **Linear time complexity:** Retrieval code iterates through all KB entries for every query — $O(n)$ time complexity. Production systems use vector databases with approximate nearest neighbor search ($O(\log n)$).

Reflection

This lab demonstrates the core RAG architecture that powers modern enterprise AI systems like Microsoft Copilot, GitHub Copilot, and domain-specific chatbots. The separation of retrieval and generation phases solves the hallucination problem by grounding LLM outputs in verified knowledge base content. However, the keyword-based retrieval approach used here has significant limitations compared to vector-based semantic search. In production RAG systems, text is converted into high-dimensional embeddings using models like text-embedding-3 or sentence-transformers, and stored in specialised vector databases like Pinecone, Weaviate, or pgvector. Retrieval then uses cosine similarity to find semantically similar content rather than exact keyword matches. This enables RAG systems to understand that "reduce costs" and "optimise expenses" are conceptually equivalent, dramatically improving retrieval accuracy. Additionally, production systems require chunking strategies to split large documents into retrievable segments, caching layers to reduce latency and API costs, and monitoring infrastructure to detect retrieval quality degradation over time. The n8n visual workflow approach enabled rapid prototyping without writing server code, but a production RAG system serving real users would require a proper backend framework, a vector database, and robust error handling and observability. This lab has provided hands-on experience with the RAG pattern that is directly applicable to research in financial AI, fraud detection systems, regulatory compliance technology, and any domain requiring AI systems grounded in private knowledge bases.

***** End Lab 04 *****